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(ACTC) ADVANCED CHEMISTRY TUITION CENTRE, 41/1 PWD ROAD, NAGERCOIL, 9940847892

+2 CHEMISTRY QUARTERLY EXAM 2025

Instructions: 1) check the question paper for fairness of printing. If there is any lack of fairness inform the Hall supervisor immediately.

2) Use **Blue or Black ink** to write and underline and pencil to draw diagrams.

- **Draw** diagrams and write equations wherever necessary**.**
- **Choose** the most appropriate answer from the given four alternatives and write the option code and corresponding answer**.**

PART- I Answer All the Questions. Choose the best Answer: 15x1=15

1. The rate constant of a reaction is 5.8 $\times 10^{-2}$ s^{-1}. The order of the reaction is
a) Second order b) first order c) Third order d) Zero order

2. In acid medium, potassium permanganate oxidizes oxalic acid to
a) acetate b) oxalate c) acetic acid d) carbon dioxide

3. Among the following which will not be hydrolysed?
a) Sodium chloride b) Sodium Formate c) Ammonium Formate d) Ammonium Nitrate

4. In the electrolytic refining of copper, which one of the following is used as anode?
a) Carbon rod b) Pure copper c) Platinum electrode d) Impure copper

5. Assertion: Monoclinic sulphur is an example of monoclinic crystal system.
Reason: For a monoclinic system, a≠b ≠c and $\alpha = \gamma = 90^0$, $\beta \neq 90^0$
a) Assertion is true but Reason is false.
b) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
c) Both Assertion and Reason are false.
d) Both Assertion and Reason are not true, but Reason is not the correct explanation of Assertion.

6. The formation of cyanohydrin from acetone is an example of:
a) electrophilic addition b) nucleophilic substitution c) nucleophilic addition d) electrophilic addition

7. Which of the following is not sp^2 hybridised?
a) Fullerene b) Graphite c) Dry ice d) Graphene

8. The oxidizing agent used to stop the oxidation of primary alcohol at the aldehyde stage is:
a) $Na_2Cr_2O_7$ b) $KMnO_4$ c) $K_2Cr_2O_7$ d) PCC

9. Which of the following is the strongest acid among all?
a) HBr b)HI c) HCl d) HF

10. Which one of the following ores is best concentrated by froth floatation method?
(a) Magnetite (b) Haematite (c) Galena (d)cassiterite

11. The element that does not show catenation among the following p-block elements is:
a) Carbon (b) Silicon (c) Lead (d) Germanium

12. The vacant space in bcc lattice unit cell is:
(a) 48% (b) 23% (c) 32% (d) 26%

13. The decomposition of phosphine (PH_3) on tungsten at low pressure is a first order reaction.
It is because the: (a) Rate is proportional to the surface coverage.
b)Rate is inversely proportional to the surface coverage.
c) Rate is independent of the surface coverage.
d) Rate of decomposition is slow.

14. The actinoid elements which show the highest oxidation state of +7 are:
(a) Np, Pu, Am (b) U, Fm, Th (c) U, Th, Md (d) Es, No, Lr

15. On hydrolysis, PCl_3 gives:
(a) H_3PO_3 (b) PH_3 (c) H_3PO_4 (d)$POCl_3$

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PART-II ANSWER ANY SIX QUESTION.QUESTION NO.24 IS COMPULSORY **6x2=12**

16. What is Calcination ?
17. How will you convert boric acid to boron nitride ?
18. Sulphuric acid is a dehydrating agent. Justify with an example.
19. What is an order of a reaction?
20. Explain common ion effect with an example.
21. Write Kolbs reaction.
22. How is Benzoic acid prepared from toluene?
23. Write Williamson ether synthesis.
24. Calculate the concentration of OH^- in a fruit juice which contains $2 \times 10^{-3}M$, H_3O^+ ion. Identify the nature of the solution. (8)

PART-III ANSWER ANY SIX QUESTION.QUESTION NO.33 IS COMPULSORY **6x3=18**

25. Give the uses of Helium.
26. What is more stable Fe^{3+} or Fe^{2+}? why?
27. what are interstitial compounds? Give an example.
28. Aluminium crystallize in a cubic close paced structure. It is metallic radius is 125pm. Calculated the edge length of unit cell.
29. Write Arrhenius equation and explain the term involved.
30. What are Lewis acids and bases?
31. Write note on Lucas test.
32. Explain Knoevenagal reaction.
33. Identify A and B.

$$\text{Benzoic acid} \xrightarrow{PCl5} A \xrightarrow{benzene/\ Anhydrous\ AlCl3} B$$

PART-IV ANSWER ALL THE QUESTIONS. **5x5=25**

34. (a) Explain the electrometallurgy of aluminium. (or)

(b) (i) What are the interhalogen compounds ? Give example.

(ii)Write the molecular formula and structural formula for the following.

(A) Phosphoric acid (B) Nitric acid

35. (a) (i) What are the differences between minerals and ores ?

(ii) What is the role of silica in the extraction of copper ? OR

(b) What is Lanthanoid Concentration and what are the consequence of Lanthanoid Contraction?

36. (a) Calculate the percentage efficiency of packing in case of simple cubic crystal. OR

(b) (i) Drive the integrated rate law for a Zero order reaction, A→ product.

(ii) Define buffer Index.

37. (a) Derive Henderson-Hasselbalch equation. OR

(b) How will you convert benzaldehyde into the following compounds?

(i) Benzoin (ii) Cinnamic acid (iii) Malachite green

38. (a) (i) How aectic acid is prepred from Grignard reagents ?

(ii) How is phenol prepared from the following?

(i) Chloro benzene (ii) Isopropyl benzene OR

(b) An organic compound (A) of molecular formula C_7H_6O undergoes Cannizaro reaction compound (A) also react with chlorine in the presence of conc.$FeCl_3$ to give compound (B) compound (A) also react with chlorine in the absence of catalyst to give compound (C). Identify A, B and C with suitable reactions.

ALL THE BEST

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